

Arrays

LEVEL 5: Two-Pointer & Sliding Window Basics

Program 61: Find Pair with Given Sum (Two Pointers) Write a program that finds if there exist two elements whose sum equals target. Example: [1,2,3,4,5], target=7 → Pair exists (2,5) or (3,4).

Program 62: Count Pairs with Given Sum Write a program that counts how many pairs have sum equal to target.

Program 63: Print All Pairs with Given Sum Write a program that displays all pairs whose sum equals target value.

Program 64: Find Triplet with Given Sum Write a program that finds three elements whose sum equals target.

Program 65: Count Pairs with Difference K Write a program that counts pairs where absolute difference is K. Example: [1,3,5,7], K=2 → Pairs: (1,3), (3,5), (5,7).

Program 66: Maximum Sum of K Consecutive Elements Write a program that finds maximum sum of K consecutive elements. Example: [1,2,3,4,5], K=3 → Max: 12 (3+4+5).

Program 67: Minimum Sum of K Consecutive Elements Write a program that finds minimum sum of K consecutive elements.

Program 68: First Negative in Every Window of Size K Write a program that finds first negative number in every window of size K.

Program 69: Maximum in Every Subarray of Size K Write a program that finds maximum element in every window of size K.

Program 70: Count Subarrays with Sum Equal to K Write a program that counts how many subarrays have sum equal to K.

Program 71: Find Subarray with Given Sum (Positive Numbers) Write a program that finds starting and ending indices of subarray with given sum. Example: [1,2,3,7,5], sum=12 → Indices: 1 to 3.

Program 72: Longest Subarray with Sum K Write a program that finds length of longest subarray with sum equal to K.

LEVEL 6: Advanced Logic & Problem Solving

Program 73: Maximum Sum Subarray (Kadane's Algorithm) Write a program that finds maximum sum of any contiguous subarray. Example: $[-2, 1, -3, 4, -1, 2, 1, -5, 4] \rightarrow \text{Max: } 6$ (subarray $[4, -1, 2, 1]$).

Program 74: Maximum Product Subarray Write a program that finds maximum product of any contiguous subarray. Example: $[2, 3, -2, 4] \rightarrow \text{Max: } 6$ (subarray $[2, 3]$).

Program 75: Equilibrium Index Write a program that finds index where sum of left elements equals sum of right elements. Example: $[1, 3, 5, 2, 2] \rightarrow \text{Index: } 2$ ($1+3 = 2+2$).

Program 76: Leader Elements in Array Write a program that finds all leader elements (element greater than all elements to its right). Example: $[16, 17, 4, 3, 5, 2] \rightarrow \text{Leaders: } 17, 5, 2$.

Program 77: Peak Element Write a program that finds a peak element (greater than or equal to its neighbors). Example: $[1, 3, 20, 4, 1, 0] \rightarrow \text{Peak: } 20$.

Program 78: Find All Peaks Write a program that finds all peak elements in the array.

Program 79: Valley Element Write a program that finds valley element (smaller than both neighbors).

Program 80: Missing Number (1 to N) Write a program where array contains numbers 1 to N with one missing. Find missing number. Example: $[1, 2, 4, 5, 6] \rightarrow \text{Missing: } 3$.

Program 81: Find Duplicate Number Write a program where array contains 1 to N with one number appearing twice. Find the duplicate.

Program 82: Find Two Missing Numbers Write a program where array contains 1 to N with two numbers missing. Find both.

Program 83: Majority Element (Boyer-Moore Algorithm) Write a program that finds element appearing more than $N/2$ times. Example: [3,3,4,2,4,4,2,4,4] → Majority: 4.

Program 84: Stock Buy and Sell (Single Transaction) Write a program where array represents stock prices. Find maximum profit by buying once and selling once. Example: [7,1,5,3,6,4] → Buy at 1, sell at 6 → Profit: 5.

Program 85: Stock Buy and Sell (Multiple Transactions) Write a program that finds maximum profit with unlimited transactions (buy-sell-buy-sell...).

Program 86: Rearrange Positive and Negative Numbers Write a program that rearranges array with alternating positive and negative numbers.

Program 87: Rearrange Array in Zig-Zag Pattern Write a program that rearranges in zig-zag: $a < b > c < d > e$. Example: [4,3,7,8,6,2,1] → [3,7,4,8,2,6,1].

Program 88: Dutch National Flag (Sort 0s, 1s, 2s) Write a program that sorts array containing only 0s, 1s, 2s in single pass without using sorting. Example: [0,1,2,0,1,2,1,0] → [0,0,0,1,1,1,2,2].

Program 89: Trapping Rainwater Write a program that calculates water trapped between array elements (heights). Example: [0,1,0,2,1,0,1,3,2,1,2,1] → Water: 6 units.

Program 90: Container With Most Water Write a program that finds two lines forming container with maximum water area. Example: [1,8,6,2,5,4,8,3,7] → Max area: 49.

Program 91: Maximum Difference (arr[j] - arr[i] where $j > i$) Write a program that finds maximum difference where larger element appears after smaller. Example: [2,3,10,6,4,8,1] → Max diff: 8 (10-2).

Program 92: Smallest Missing Positive Number Write a program that finds smallest missing positive integer. Example: [3,4,-1,1] → Missing: 2.

LEVEL 7: Sorting Algorithms

Program 93: Bubble Sort Write a program that sorts array using bubble sort (repeatedly swap adjacent elements if wrong order).

Program 94: Selection Sort Write a program that sorts using selection sort (repeatedly find minimum and place at beginning).

Program 95: Insertion Sort Write a program that sorts using insertion sort (build sorted array one element at a time).

Program 96: Count Sort (Limited Range 0-9) Write a program that sorts array of digits (0-9) using counting sort.

Program 97: Sort Array of 0s and 1s Write a program that sorts binary array (only 0s and 1s) in single traversal.

Program 98: Sort Array of 0s, 1s, and 2s (Dutch Flag) Write a program that sorts array with three distinct values in single pass.

Program 99: Merge Two Sorted Arrays Write a program that merges two sorted arrays into third sorted array without using extra sorting.

Program 100: Check if Array Can Become Sorted by Single Swap Write a program that checks if array can be sorted by swapping exactly two elements once.

LEVEL 8: Binary Search & Advanced Searching

Program 101: Binary Search (Iterative) Write a program that searches element in sorted array using binary search iteratively.

Program 102: Binary Search (Recursive) Write a program that implements binary search using recursion.

Program 103: First Occurrence in Sorted Array Write a program that finds first position of element in sorted array with duplicates.

Program 104: Last Occurrence in Sorted Array Write a program that finds last position of element in sorted array with duplicates.

Program 105: Count Frequency in Sorted Array Write a program that counts frequency of element using binary search (last - first + 1).

Program 106: Search in Rotated Sorted Array Write a program that searches in rotated sorted array. Example: [4,5,6,7,0,1,2], find 0.

Program 107: Find Minimum in Rotated Sorted Array Write a program that finds minimum element in rotated sorted array.

Program 108: Find Rotation Count Write a program that finds how many times array was rotated. Example: [4,5,6,7,0,1,2] → Rotated 4 times.

Program 109: Search Insert Position Write a program that finds index where element should be inserted to maintain sorted order.

Program 110: Find Peak Element Using Binary Search Write a program that finds peak element efficiently using binary search.

Program 111: Square Root Using Binary Search Write a program that finds integer square root using binary search approach.

Program 112: Find Floor and Ceiling in Sorted Array Write a program that finds floor (largest element $\leq X$) and ceiling (smallest element $\geq X$).

Program 113: Count Elements Greater Than X in Sorted Array Write a program that counts elements greater than X using binary search.

Program 114: Find Kth Smallest Element Using Binary Search Write a program that finds Kth smallest using binary search on answer approach.

Program 115: Median of Two Sorted Arrays Write a program that finds median when two sorted arrays are merged (without actually merging).

LEVEL 9: Advanced Array Challenges

Program 116: Longest Consecutive Sequence Write a program that finds length of longest consecutive elements. Example: [100,4,200,1,3,2] → Length: 4 (1,2,3,4).

Program 117: Product of Array Except Self Write a program that creates array where each element is product of all others except itself (without division). Example: [1,2,3,4] → [24,12,8,6].

Program 118: Shuffle Array Write a program that randomly shuffles array elements (Fisher-Yates algorithm).

Program 119: Next Permutation Write a program that rearranges array into next lexicographically greater permutation. Example: [1,2,3] → [1,3,2].

Program 120: Previous Permutation Write a program that finds previous lexicographically smaller permutation.

Program 121: Subarray with Maximum XOR Write a program that finds maximum XOR of any subarray.

Program 122: Count Inversions Write a program that counts inversions (pairs where $i < j$ but $\text{arr}[i] > \text{arr}[j]$). Example: $[8,4,2,1] \rightarrow$ Inversions: 6.

Program 123: Find All Triplets with Sum Zero Write a program that finds all unique triplets that sum to zero. Example: $[-1,0,1,2,-1,-4] \rightarrow$ Triplets: $[[-1,-1,2],[-1,0,1]]$.

Program 124: Longest Increasing Subsequence Length Write a program that finds length of longest increasing subsequence. Example: $[10,9,2,5,3,7,101,18] \rightarrow$ LIS: 4 ($[2,3,7,101]$).

Program 125: Maximum Circular Subarray Sum Write a program that finds maximum sum considering array as circular. Example: $[8,-8,9,-9,10,-11,12] \rightarrow$ Max: 22 ($12+8-8+9-9+10$).